

EXPERIMENT 08

User Management and Role-Based Access Control (RBAC) in Zoho Creator

Permissions, Roles, and Data Sharing for the Loan Management System

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Aim

To explore user management and role-based access control (RBAC) in Zoho Creator by reviewing the Permissions, Roles and Data Sharing settings of the Loan Management System, and to understand how these three layers decide who can see and change which loan records.

Objectives

- To review the Zoho Creator home page and the application dashboard as the entry points to the Loan Management System.
- To study the permission sets defined for the application (Branch Manager, Credit Manager, Customer Service Executive, Loan Officer, Read, Write).
- To build and review the role hierarchy, from the CEO down to the operational roles.
- To create a data sharing rule that lets one role access records submitted by another.
- To save the rule and verify that every field was stored exactly as configured.

Software / Tools Required

- Zoho Creator (web-based low-code application development platform)
- Loan Management System application
- Web browser with an active internet connection
- Zoho account with access to the application in Edit mode

Introduction / Theory

Role-Based Access Control in Zoho Creator

Role-based access control (RBAC) grants access according to the job a person does rather than to the individual. In Zoho Creator, access to an application is controlled in three layers that work together: permissions decide what a user may do with forms, reports and pages; roles arrange users in a hierarchy that decides whose records a user can see; and data sharing rules open specific records to other roles or users beyond what the hierarchy already allows.

Permissions, Roles and Data Sharing

A permission set (profile) is a named bundle of rights, such as the right to view, add, edit or delete records in selected forms. A role is a position in the organisation tree; a user with a role can normally see the records of the roles beneath it. A data sharing rule is an explicit exception: it names a form, the role or users whose records are shared, the role or users they are shared with, and the level of access (Read or Read/Write). Together they let an application show each user only the loan data their work requires.

Key Concepts

Concept	Description
Permission Set	A named bundle of rights that defines what a user can do with the application's forms, reports and pages.
Role	A position in the role hierarchy that decides whose records a user can see.
Role Hierarchy	A tree of roles in which each role can see the records of the roles beneath it.
Data Sharing Rule	A rule that shares records of a chosen form from one role or user group with another, at Read or Read/Write access.
Level of Access	The degree of access granted by a share rule: Read allows viewing, Read/Write allows viewing and editing.

Application Specifications

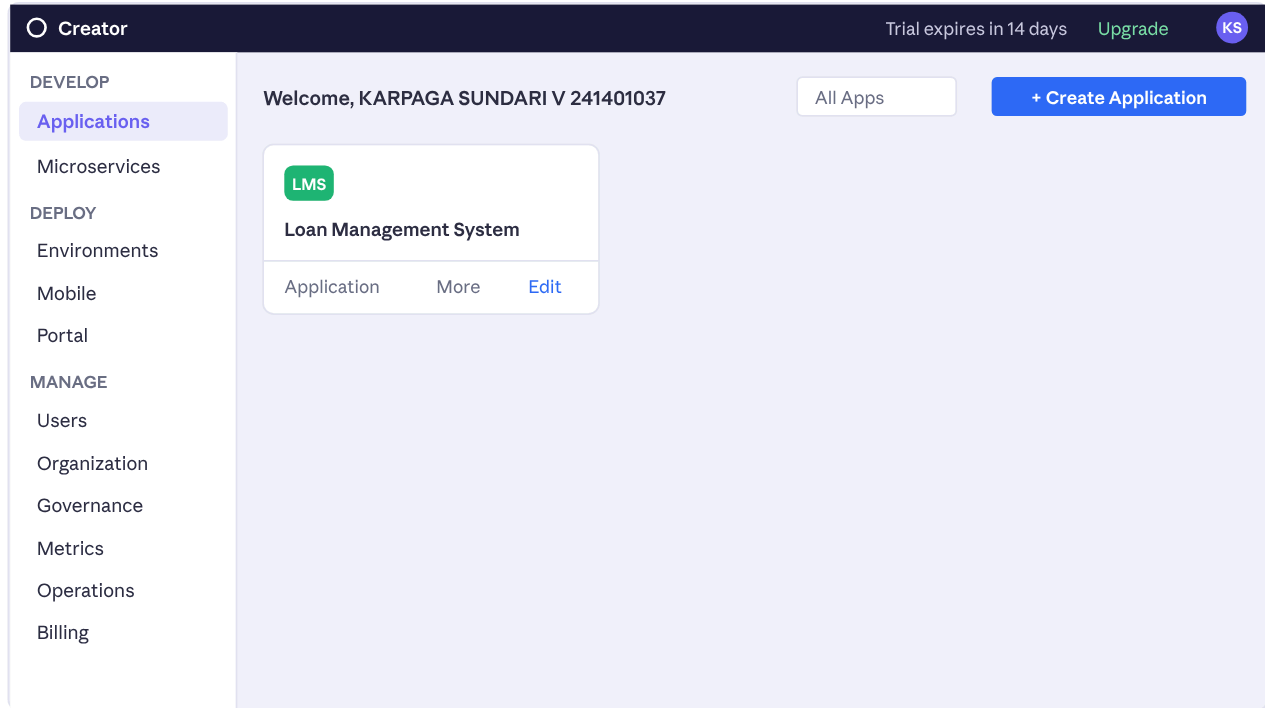
Parameter	Detail
Application Name	Loan Management System
Primary Forms	Borrower, Loan Application, Loan Type, Repayment
Permission Sets	Branch Manager, Credit Manager, Customer Service Executive, Loan Officer, Read, Write
Role Hierarchy	CEO → Branch Manager → Credit Manager, Loan Officer, Customer Service Executive
Data Sharing Rule	Customer_Service_to_Branch_Manager on the Loan Application form (Read/Write)

Procedure

The figures in this record are illustrative representations drawn with sample values; they are not screenshots of the application.

1. Open the Zoho Creator home page - Signing in to Zoho Creator opens the Home page, which lists the applications in the account.

The home page lists the Loan Management System application



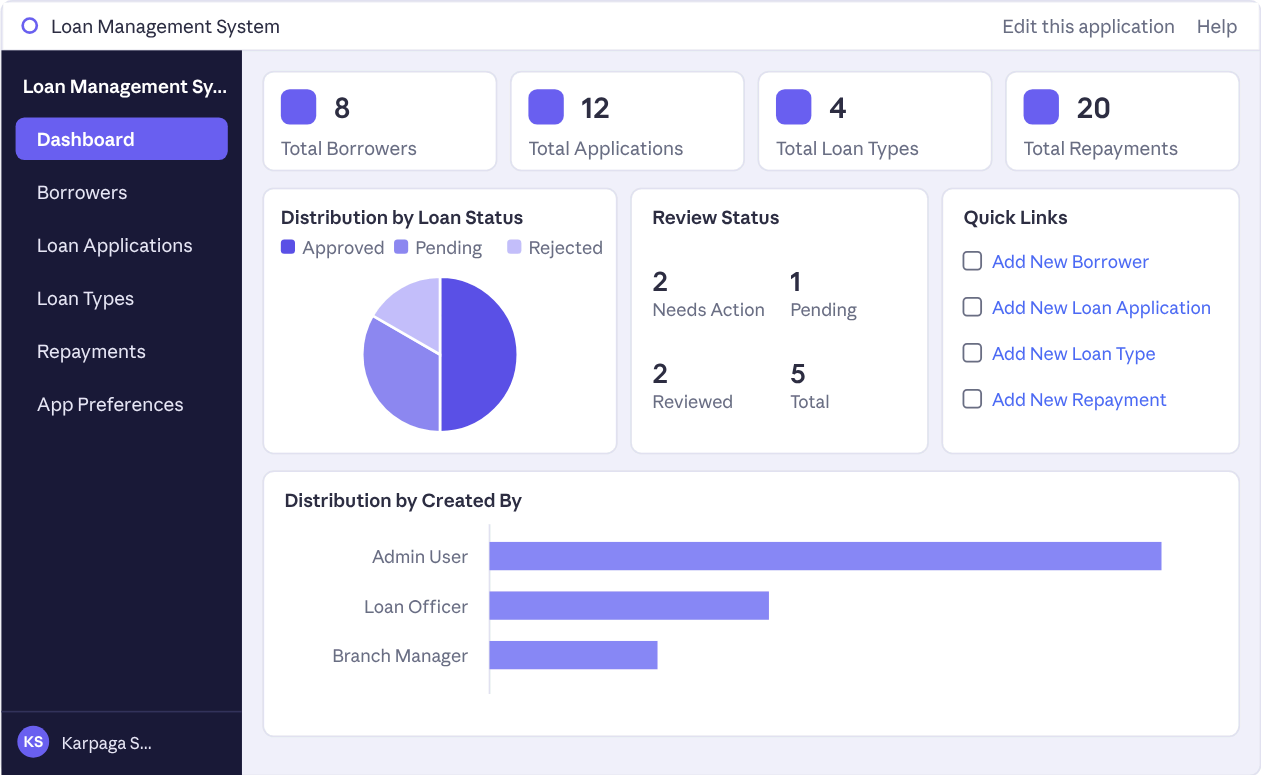
Zoho Creator home · application card

Figure 8.1: Zoho Creator Home - Loan Management System (Illustrative Representation, Not a Screenshot)

The home page lists the Loan Management System as an application card, alongside the Application and Edit shortcuts.

2. Open the dashboard - Opening the application loads its Dashboard, showing summary tiles and charts built from the loan data.

The dashboard shows loan data in tiles and charts



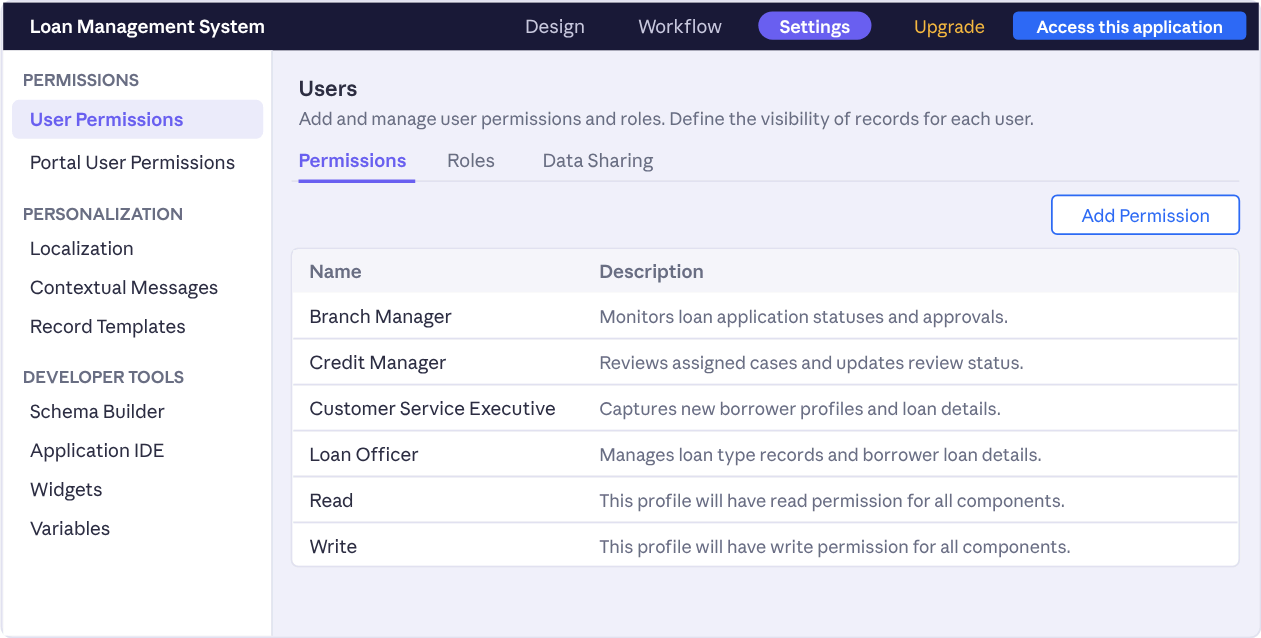
Dashboard in Access mode (sample values)

Figure 8.2: Loan Management System - Dashboard (Access Mode) (Illustrative Representation, Not a Screenshot)

The dashboard shows four summary tiles (Borrowers, Loan Applications, Loan Types, Repayments), a Distribution by Loan Status pie chart, Review Status counts, Quick Links, and a Distribution by Created By bar chart.

- 3. Enter Edit mode and open Settings - Selecting Edit this application switches to Edit mode, and the Settings tab is selected from the top navigation bar (Design, Workflow, Settings).
- 4. Open User Permissions - Under the PERMISSIONS group, User Permissions is selected, opening the Users page with the Permissions, Roles and Data Sharing tabs.

Six permission sets define what each kind of user can do



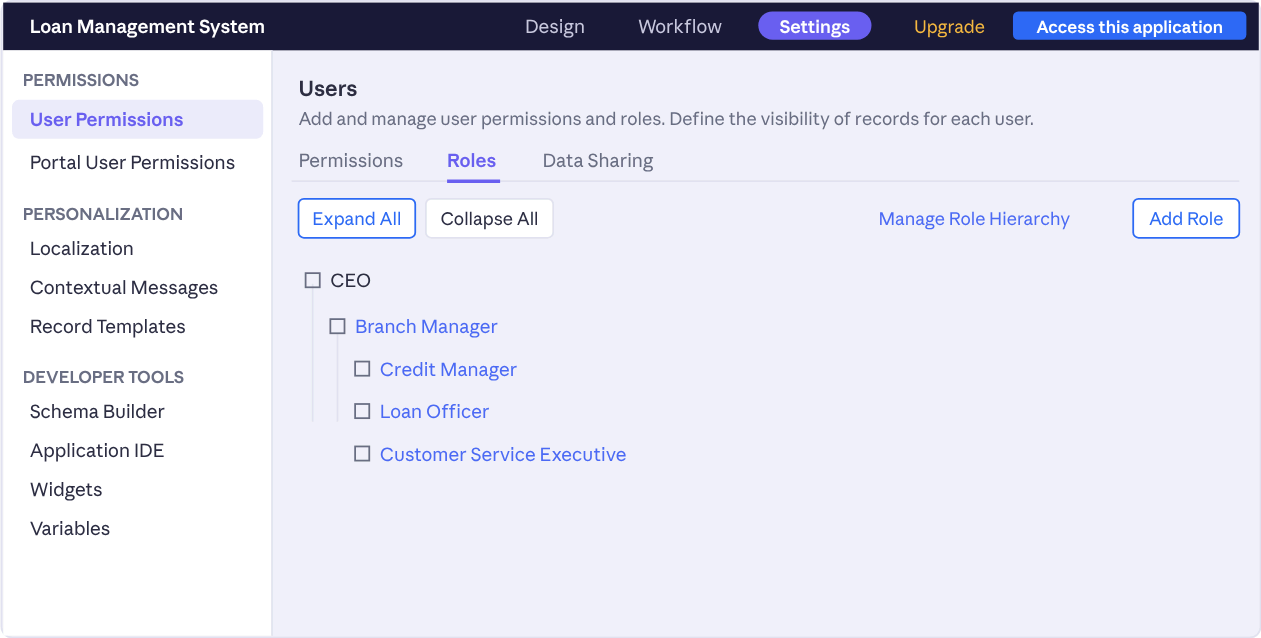
Settings · Permissions tab · six permission sets

Figure 8.3: User Permissions - Permissions Section (Illustrative Representation, Not a Screenshot)

The Permissions tab lists six permission sets: Branch Manager (monitors loan application statuses including pending, approved and rejected), Credit Manager (reviews assigned applications and updates review status as required), Customer Service Executive (captures new borrower profiles and initial loan details), Loan Officer (manages loan type records and borrower loan details), Read, and Write, each with a short description and an Add Permission option.

5. Build the role hierarchy - On the Roles tab, the role tree is arranged as CEO → Branch Manager → Credit Manager, Loan Officer, and Customer Service Executive, using Expand All, Collapse All, Manage Role Hierarchy and Add Role as needed.

Roles form a tree: CEO, Branch Manager, then three operational roles



Settings · Roles tab · role hierarchy

Figure 8.4: Roles and Role Hierarchy (Illustrative Representation, Not a Screenshot)

The Roles tab shows the completed hierarchy: CEO at the top, Branch Manager reporting to it, and Credit Manager, Loan Officer and Customer Service Executive all reporting to Branch Manager as sibling roles.

6. Create a new share rule - On the Data Sharing tab, New Share Rule is selected, the rule is named Customer_Service_to_Branch_Manager, the Form is set to Loan Application, and Records shared from is set to Roles → Customer Service Executive.

A new share rule shares Loan Application records between two roles

Loan Management System

Design Workflow

PERMISSIONS

User Permissions

Portal User Permissions

PERSONALIZATION

Localization

Contextual Messages

Record Templates

Users

Permissions Roles **Data Sharing**

You have not added any data sharing rules yet

Add Rule

New Share Rule

Name

Customer_Service_to_Branch_Manager

Form

Loan Application

Records shared from

Roles

Customer Service Executive

Records shared To

Roles

Branch Manager

Level of access

Read/Write

Save Cancel

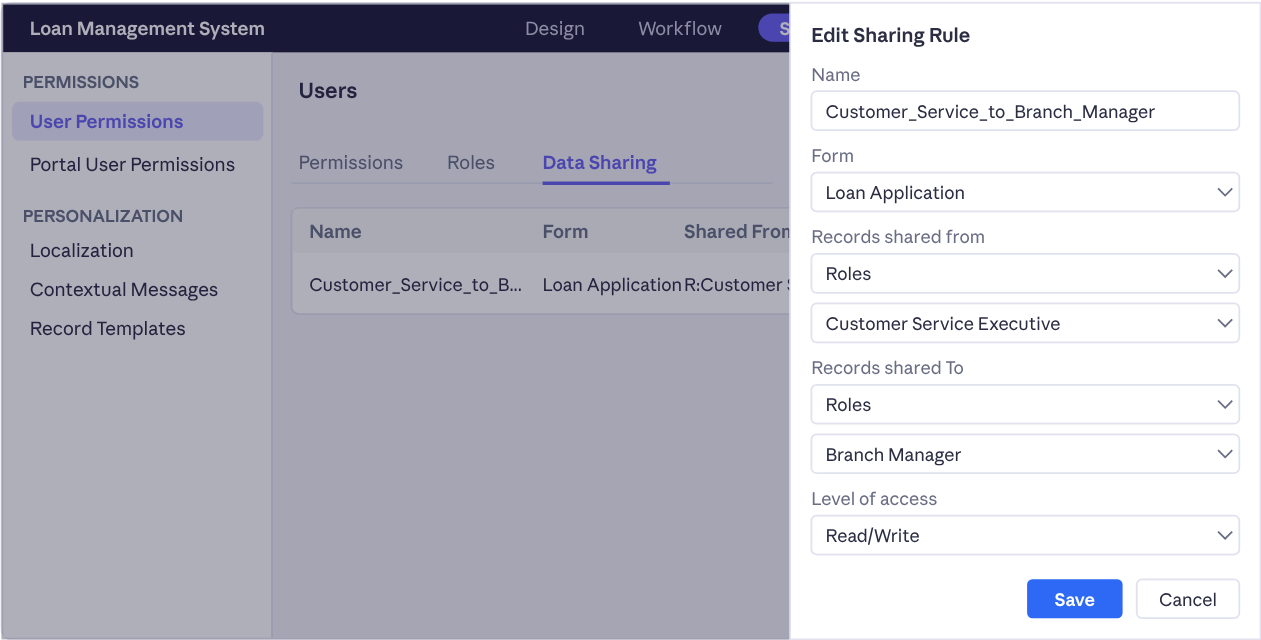
Settings · Data Sharing tab · New Share Rule panel

Figure 8.5: New Share Rule - Rule Configuration (Illustrative Representation, Not a Screenshot)

The New Share Rule panel shows the rule `Customer_Service_to_Branch_Manager` on the Loan Application form, with Records shared from set to Roles → Customer Service Executive, Records shared To set to Roles → Branch Manager, and Level of access set to Read/Write.

7. Save and verify the rule - Selecting Save adds the rule to the Data Sharing list; reopening it with Edit Sharing Rule confirms every field was stored exactly as configured.

The saved rule appears in the Data Sharing list with Read/Write access



Settings · Data Sharing list · Edit Sharing Rule panel

Figure 8.6: Configured Data Sharing Rule (Customer Service Executive → Branch Manager, Read/Write) (Illustrative Representation, Not a Screenshot)

The Data Sharing list now shows the saved rule: Form: Loan Application, Shared From: R:Customer Service Executive, Shared To: R:Branch Manager, and the Edit Sharing Rule panel confirms the Read/Write access level.

Data Sharing

The application follows a layered model: Form (Borrower / Loan Application / Loan Type / Repayment) → Dashboard → Security Layer (Permissions, Roles, Data Sharing). The role hierarchy runs from the CEO (overall oversight) down through Branch Manager to Credit Manager, Loan Officer and Customer Service Executive, so each role sees the records of the roles beneath it.

The configured share rule adds one exception to this pattern. Loan applications captured by the Customer Service Executive role are shared with the Branch Manager role at Read/Write access, so the branch can review and update them directly.

Parameter	Detail
Rule Name	Customer_Service_to_Branch_Manager
Form	Loan Application
Records Shared From	Roles → Customer Service Executive
Records Shared To	Roles → Branch Manager
Level of Access	Read/Write

Result

- The Zoho Creator home page and the Loan Management System dashboard were reviewed as the entry points to the application.
- Six permission sets (Branch Manager, Credit Manager, Customer Service Executive, Loan Officer, Read, Write) were reviewed on the Permissions tab.
- The role hierarchy was arranged as CEO → Branch Manager → Credit Manager, Loan Officer and Customer Service Executive.
- A data sharing rule, Customer_Service_to_Branch_Manager, was created on the Loan Application form with Read/Write access, saved, and verified by reopening it.

Conclusion

The experiment demonstrated how Zoho Creator controls access through three cooperating layers. Permission sets define what each kind of user may do, the role hierarchy decides whose records each role can see, and data sharing rules open selected records to other roles when the hierarchy alone is not enough. Configuring the rule through the form and verifying it by reopening showed that access can be set up without writing code, which keeps loan data visible to the people who need it and hidden from everyone else.